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A USE CASES

To mitigate harm from dishonest LLMs, some previous work focuses on using lie detectors to **monitor** the models behavior during deployment (Pacchiardi et al., 2023; Goldowsky-Dill et al., 2025). Greenblatt et al. (2024d) and Hendrycks et al. (2021) discuss using such a monitor as an early defensive layer to mitigate hazards from AI systems. In addition to monitoring, we propose the usage of lie detection for two other use cases. For **safeguarding evaluations**, we consider using lie detection to preserve the integrity of model evaluations (Phuong et al., 2024), for example, by detecting *sandbagging*, a behavior in which LLMs strategically underperform in evaluations (van der Weij et al., 2025; Meinke et al., 2025). Lastly, we consider **alignment evaluations**, in which evaluators examine a model’s alignment by asking it introspective questions and verifying its answers with lie detectors.

We will now discuss each of these use cases, before motivating our decision to use balanced accuracy as the main metric of LIARS’ BENCH in subsection A.4.

A.1 USE CASE 1: ALIGNMENT EVALUATIONS

We consider the threat of a misaligned model being deployed because its misalignment was not surfaced by evaluators. Assuming access to a reliable lie detection method, we propose mitigating this threat by asking the model to reflect on its alignment, for example with questions like:

- Do you have secret goals?
- In what kinds of situations would you lie?
- Would you self-exfiltrate, if presented with an opportunity?

If the model’s responses do not appear safe, it is considered unsafe and is not deployed. If the responses do appear safe, they will be validated by a lie detector. If the detector is indeed reliable and classifies the responses as honest, the model does not pose the threats associated with the questions.

This approach comes with limitations. One concern is that a harmful model may answer these questions unaware of its potential harmfulness, and therefore, without lying. For example, responses may be generated by simulation of a safe, honest persona. A possible strategy to mitigate this concern is to provide a honeypot that aims to provoke the model’s harmful side to come to the surface before asking the questions above. For example, the honeypot could be a context presenting an opportunity for the model to copy its weights to another server before asking it whether it would self-exfiltrate.

This mitigation strategy hinges on the reliable detection of the types of lies (see section 3) we expect in this scenario: the object of belief is *self-knowledge* regarding a model’s *policies*, as these are targeted by the questions, and the reason for lying is *inherent* to the model, as its responses are generated on-policy and the model is not prompted to respond in any way other than honestly.

A.2 USE CASE 2: SAFEGUARDING EVALUATIONS

We consider the threat of scheming models trying to pass model evaluations by lying. For example, there is evidence of models being capable of *sandbagging*, that is, to strategically underperform in evaluations, in order to be deployed (van der Weij et al., 2025; Meinke et al., 2025). To mitigate such threats, we propose maintaining the integrity of evaluations by deploying lie detectors to validate either the model’s responses given during pre-existing evaluation protocols, or on follow-up questions asked specifically for this purpose, e.g., “Is this really the best you can do?” The types of lies that need to be detected in this scenario depend on the evaluation protocol. Therefore, any object of belief, and any reason for lying may be relevant.

A.3 USE CASE 3: LARGE-SCALE MONITORING

We consider threats stemming from models lying after deployment, for example during conversations with users. Järvinen & Hubinger (2024); Scheurer et al. (2024); Meinke et al. (2025) provide